LAB 7

Fuzzing for Input Validation Bugs

Water Utility Billing

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Evidence: Screenshot 1 – Original application failure

The original application accepts the first two inputs, then raises a ValueError when the billing usage field contains non-numeric text (Ab:us:bng).

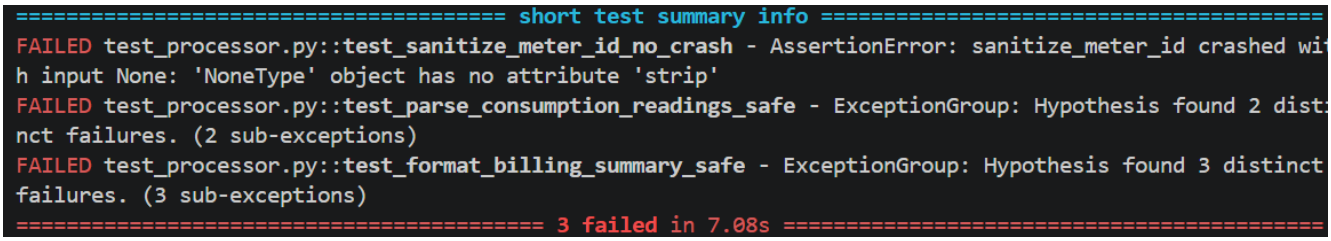
```
(venv) PS C:\Users\lenovo\OneDrive\Desktop\SE Lab\L6_L7_01_water_utility_billing> python -m pytest --vers
ion
pytest 9.2.0
(venv) PS C:\Users\lenovo\OneDrive\Desktop\SE Lab\L6_L7_01_water_utility_billing> python ".\Lab 7\process
or.py"
==== Running sanitize_meter_id ====
Enter a meter ID with prefixes (!,@,#,$,%): !123
Sanitized Meter ID: 123

==== Running parse_consumption_readings ====
Enter consumption readings (e.g. 10,25,30): 25
Parsed Readings: [25]

==== Running format_billing_summary ====
Enter account record (Name:Usage:Zone): Ab:us:bng
Traceback (most recent call last):
  File "C:\Users\lenovo\OneDrive\Desktop\SE Lab\L6_L7_01_water_utility_billing\Lab 7\processor.py", line
35, in <module>
    main()
    ~~~~^^
  File "C:\Users\lenovo\OneDrive\Desktop\SE Lab\L6_L7_01_water_utility_billing\Lab 7\processor.py", line
31, in main
    summary = format_billing_summary(summary_input)
  File "C:\Users\lenovo\OneDrive\Desktop\SE Lab\L6_L7_01_water_utility_billing\Lab 7\processor.py", line
14, in format_billing_summary
    units = int(parts[1])
ValueError: invalid literal for int() with base 10: 'us'
(venv) PS C:\Users\lenovo\OneDrive\Desktop\SE Lab\L6_L7_01_water_utility_billing> Get-Content ".\Lab 7\pr
ocessor.py"
```

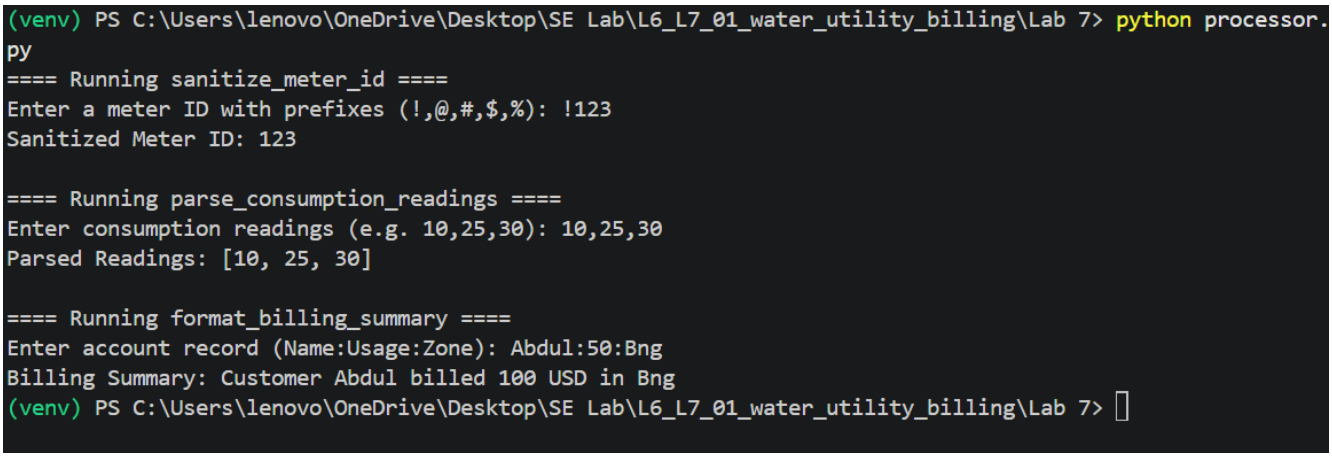
Evidence: Screenshot 2 – Hypothesis failures

The three property-based tests fail against the original implementation. Hypothesis exposes a NoneType error in meter ID sanitization and multiple invalid-input failures in the consumption and billing functions.



Evidence: Screenshot 3 – Fixed application execution

The hardened application processes valid sample inputs successfully: !123 becomes 123, 10,25,30 becomes [10, 25, 30], and Abdul:50:Bng produces a billing summary of 100 USD.



Evidence: Screenshot 4 – Passing Hypothesis tests

After the fixes, all three Hypothesis tests pass. The recorded test run reports 3 passed in 0.96 seconds.

